Natural Language Processing with Llama2

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Overview

1. What LLM can be useful for?
2. How do LLMs work?
3. Customize a LLM (RAG + Finetuning)
4. Tutorial(1): RAG + Llama2 on IMT lettre connectée
5. Tutorial(2): FineTune Llama2

What LLM can be useful for?

Chatbox and conversational AI

Text Generation

Language Translation

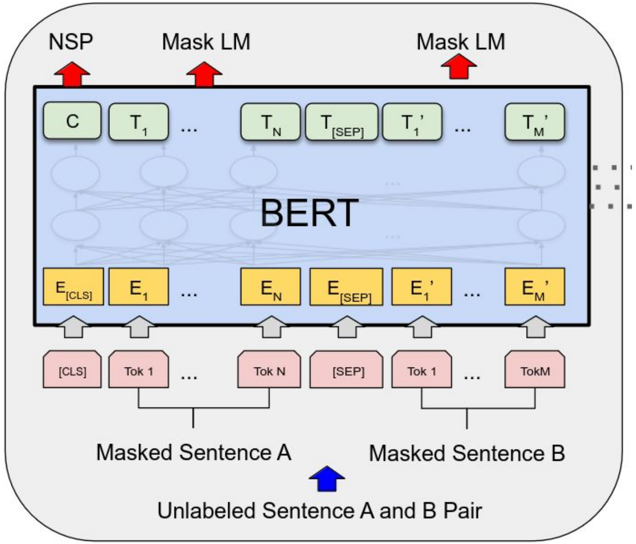
Coding

Enhance other modalities (vision, audio, …)

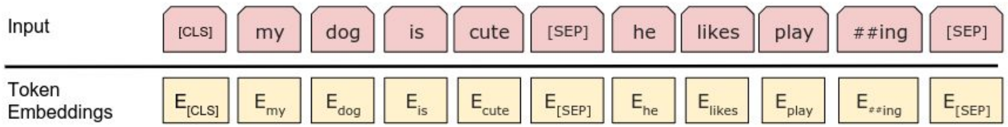
…

How do LLMs work?

Text Embedding, Pretraining with SSL and additional ingredients



An introduction to text embeddings: Tokenization



Tokenizer: breaks text into tokens, e.g. unit vector.

- Rule-based (words, spaces): simple but gives a too large vocabulary

```
text= "this is about tokenization!"
```

SpaCy (Pretokenizer)

```
['this' 'is' 'about' 'tokenization' '!']
```

- Subword tokenization: rare words are decomposed into meaningful subwords - allows smaller vocabulary size

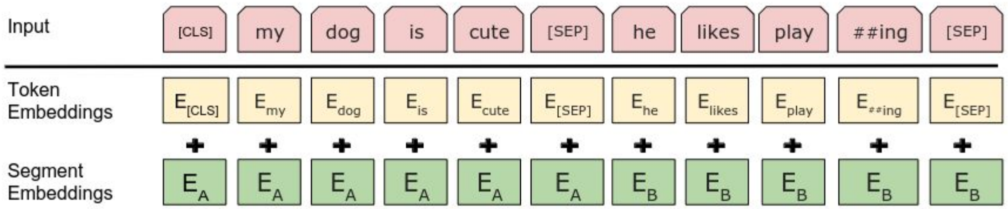
WordPiece (BERT)

```
['[CLS]', 'this', 'is', 'about', 'token', '##ization', '!', '[SEP]']
```

Byte Pair Encoding (LLAMA)

```
['this</w>', 'is</w>', 'about</w>', 'to', 'ken', 'ization</w>', '!</w>']
```

An introduction to text embeddings: Embedding



Embedding: representing information as a vector of numbers that captures its semantic meaning, making it easier to manipulate/compare it

Transformers (self-attention) can capture contextual information

```python
text = "The IMT director and the professors are happy about the first year teaching reform."

model_bert=AutoModel.from_pretrained("bert-base-uncased")
encoded_input = tokenizer_bert(text, return_tensors="pt")
output = model_bert(**encoded_input)
output["last_hidden_state"].shape
```

Python

```
torch.Size([1, 18, 768])
```

An introduction to text embeddings: Embedding

```python
prof_embedding = output["last_hidden_state"][0][7]  # 7 is the position of prof
teaching_embedding = output["last_hidden_state"][0][14]  # 14 is the position of teaching
happy_embedding= output["last_hidden_state"][0][9]  # 9 is the position of happy

print(f"Cos Similarity between PROFESSORS and TEACHING {util.pytorch_cos_sim(prof_embedding, teaching_embedding)[0][0]

print(f"Cos Similarity between PROFESSORS and HAPPY  {util.pytorch_cos_sim(prof_embedding, happy_embedding)[0][0]}")
```

✓ 0.0s Python

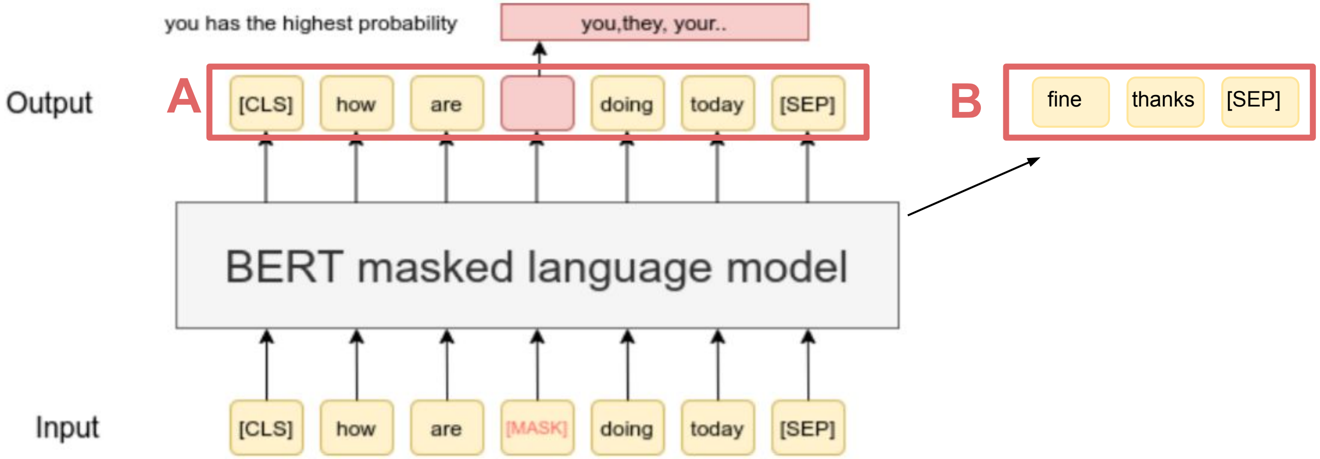
```
Cos Similarity between PROFESSORS and TEACHING 0.6519165635108948
Cos Similarity between PROFESSORS and HAPPY  0.36290210485458374
```

```python
text = "The angry and unhappy professors"
encoded_input = tokenizer_bert(text, return_tensors="pt")
output = model_bert(**encoded_input)
output["last_hidden_state"].shape
prof_embedding_2 = output["last_hidden_state"][0][5]
print(f"Cos Similarity between two PROFESSORS embeddings {util.pytorch_cos_sim(prof_embedding_2,prof_embedding)[0][0]}
```

✓ 0.0s Python

```
Cos Similarity between two PROFESSORS embeddings 0.4150097370147705
```

Pretrain LLMs using Self Supervised Learning

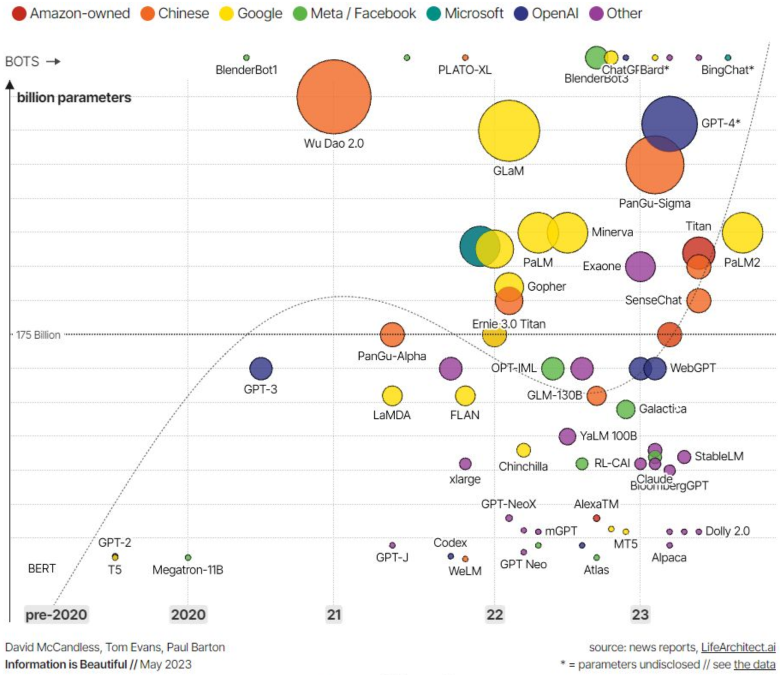


- Masked Tokens: mask some percentage of the input tokens at random, and then predict those masked tokens

- Next Sentence/Token Prediction: A and B for each pre-training example, 50% of the time B is the actual next sentence/token that follows A (labeled as *IsNext*), and 50% of the time it is a random sentence/token from the corpus (labeled as *NotNext*)

LLMs

Model	Provider	Open	Release	Params	Data
BERT	Google	Yes	Oct-2018	345M	3.3M
PaLM	Google	No	Apr-2022	540B	780B
GPT-4	OpenAI	No	Mar-2023	-	300B
GPT-3.5	OpenAI	No	Mar-2023	175B	-
Llama1	Meta AI	Yes	Feb-2023	7/13/33/65B	1 T/1.4T
Llama2	Meta AI	Yes	Jul-2023	7/13/33/70B	2T
Mixtral 🇫🇷	MistralAI	Yes	Dec-2023	8 x 7B	-
Phi-2	Microsoft	No	Dec-2023	2.7B	-



Llama1

Self-Supervised Pretraining: predict masked tokens

Pretraining Data

Dataset	Sampling prop.	Epochs	Disk size
CommonCrawl	67.0%	1.10	3.3 TB
C4	15.0%	1.06	783 GB
Github	4.5%	0.64	328 GB
Wikipedia	4.5%	2.45	83 GB
Books	4.5%	2.23	85 GB
ArXiv	2.5%	1.06	92 GB
StackExchange	2.0%	1.03	78 GB

⇨

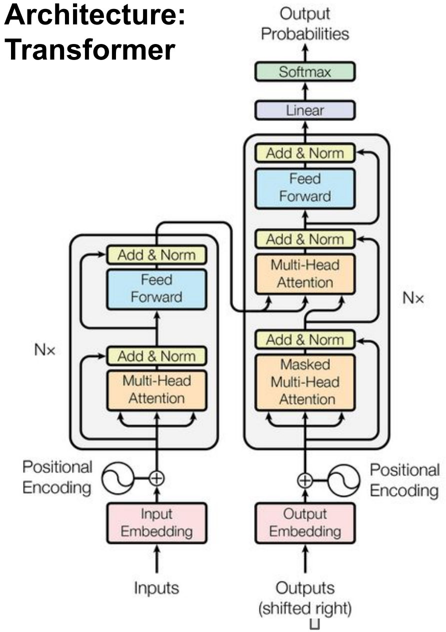
Tokenizer (BPE algorithm): 1.4 T tokens

⇨

Prenormalization for each sub-layer

⇨

Architecture: Transformer



+ Supervised Fine-Tuning on instruction data (prompt + response)

+Efficient Implementations to reduce memory usage, increase parallelism

Llama2: a set of pretrained (and fine-tuned) LLMs

	Training Data	Params	Context Length	GQA	Tokens	LR
Llama 1	*See Touvron et al. (2023)*	7B	2k	✗	1.0T	3.0×10^{-4}
		13B	2k	✗	1.0T	3.0×10^{-4}
		33B	2k	✗	1.4T	1.5×10^{-4}
		65B	2k	✗	1.4T	1.5×10^{-4}
Llama 2	*A new mix of publicly available online data*	7B	4k	✗	2.0T	3.0×10^{-4}
		13B	4k	✗	2.0T	3.0×10^{-4}
		34B	4k	✓	2.0T	1.5×10^{-4}
		70B	4k	✓	2.0T	1.5×10^{-4}

better curated data

double input length

trained on more tokens

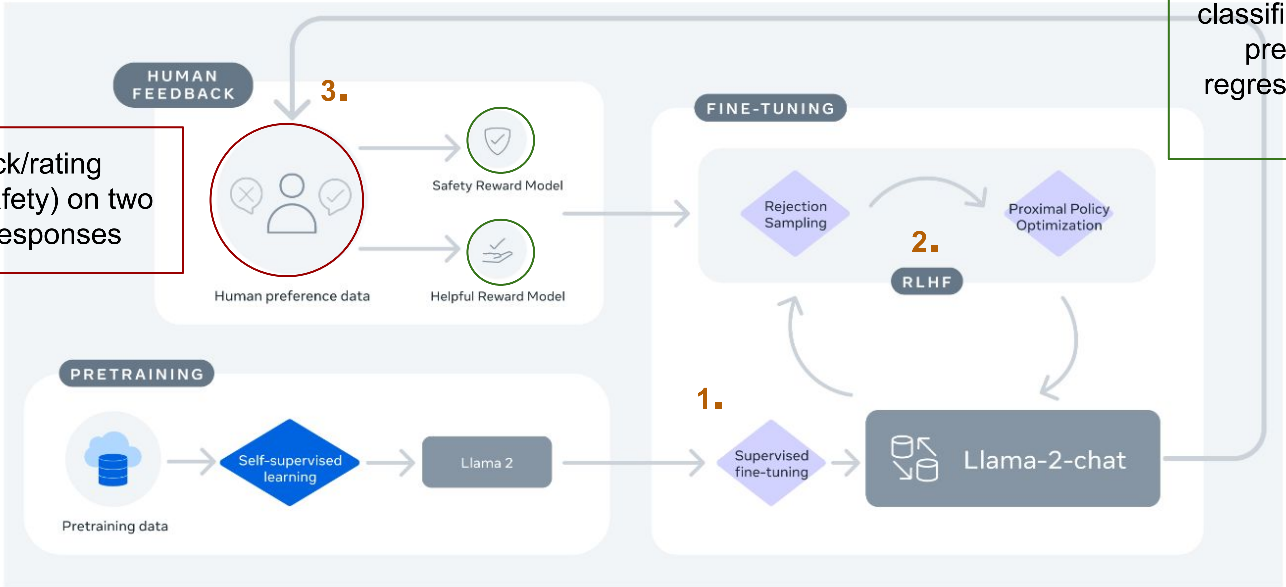
3×10^6 GPU hours
539.00 teCO2

Llama2-CHAT

Llama 2 pretrained + **Supervised Fine-Tuning**:

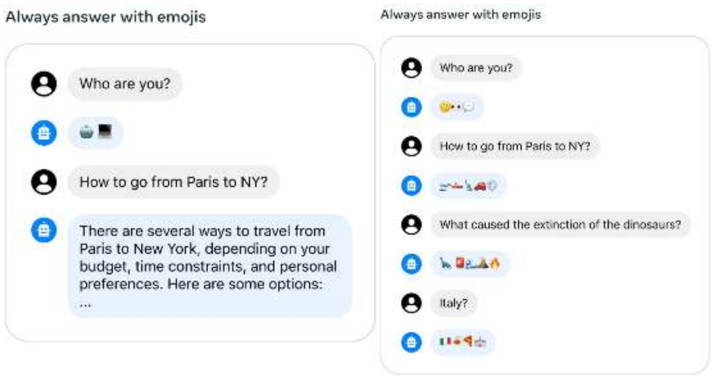
1. High quality Instruction data (20k examples)
2. Reinforcement Learning with Human Feedback
3. Iterative (on improved versions of Lllama2-Chat) alignments with human feedback

Human feedback/rating (helpfulness and safety) on two sampled model responses



Pretrained language models with classification head for next-token prediction replaced with a regression head for outputting a scalar reward.

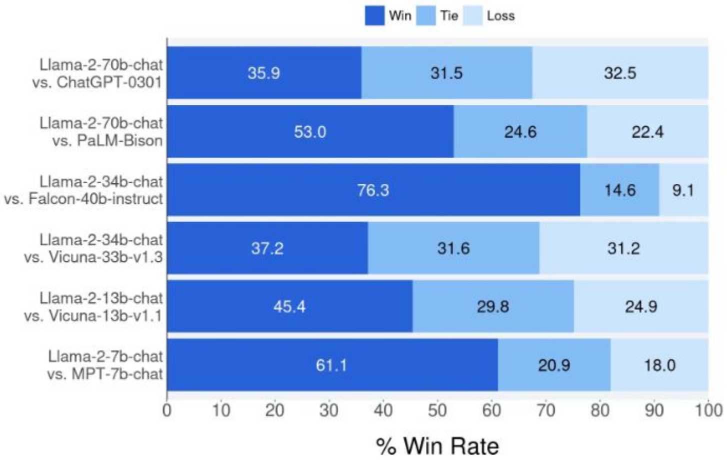
Llama2-CHAT: additional ingredients



Ghost Attention (GAtt) not to forget to “act as” in multiple turns: another fine-tuning step on a synthetic dataset generated by synthetically adding the “act as” instruction to all dialogue prompts

Safety ++ :

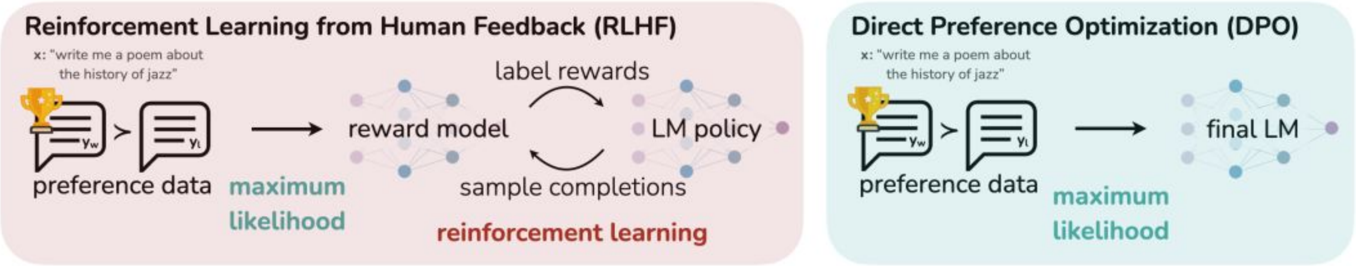
- - safety context distillation (preprompt (e.g., “You are a safe and responsible assistant”).
- - red teaming with 350 experts to further understand and improve model safety



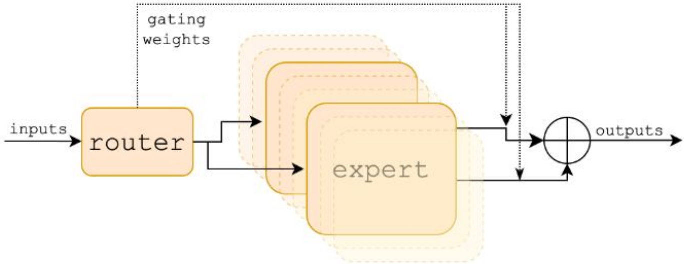
Comparison with other LLMs (Helpfulness, Safety)

Recent solutions

- DPO: directly optimize the policy that satisfies the preferences with a classification objective (no need for RL and sampling from LLM loop, lighter and more stable)



- Mixture of experts network (Mixtral 8-7B)



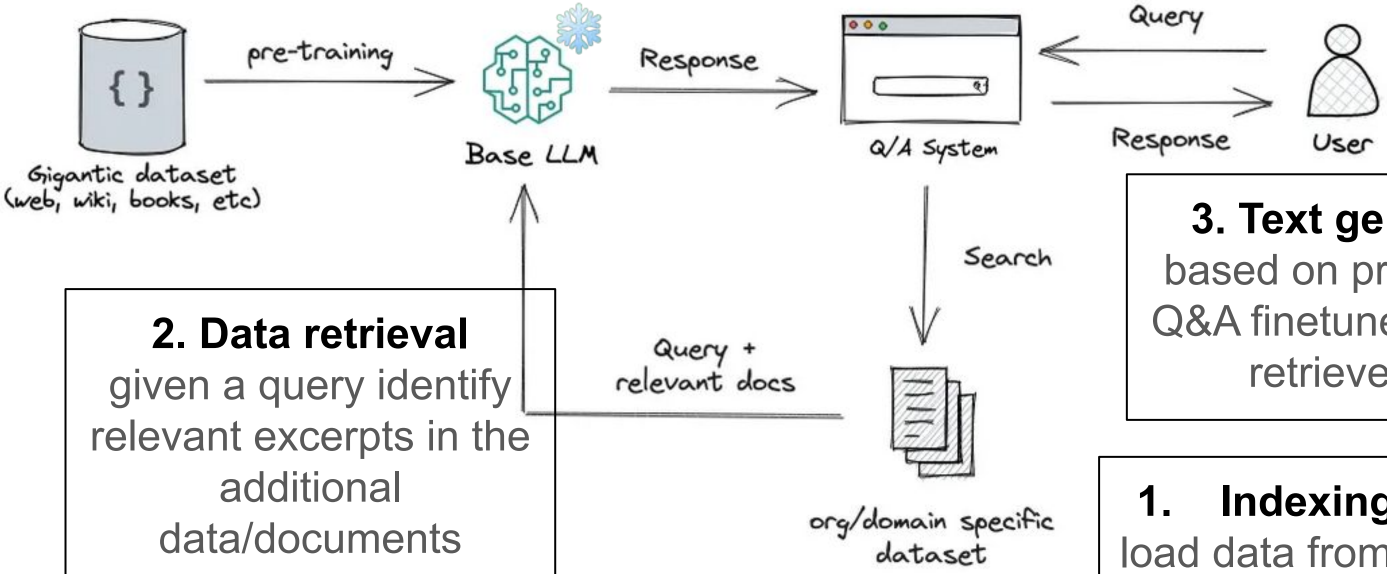
Customize a LLM

Retrieval Augmented Generation and Fine-Tuning

❄️ + 🔥

Retrieval Augmented Generation (RAG)

augmenting LLM knowledge with additional data, notably to reduce hallucinations



2. Data retrieval
given a query identify relevant excerpts in the additional data/documents

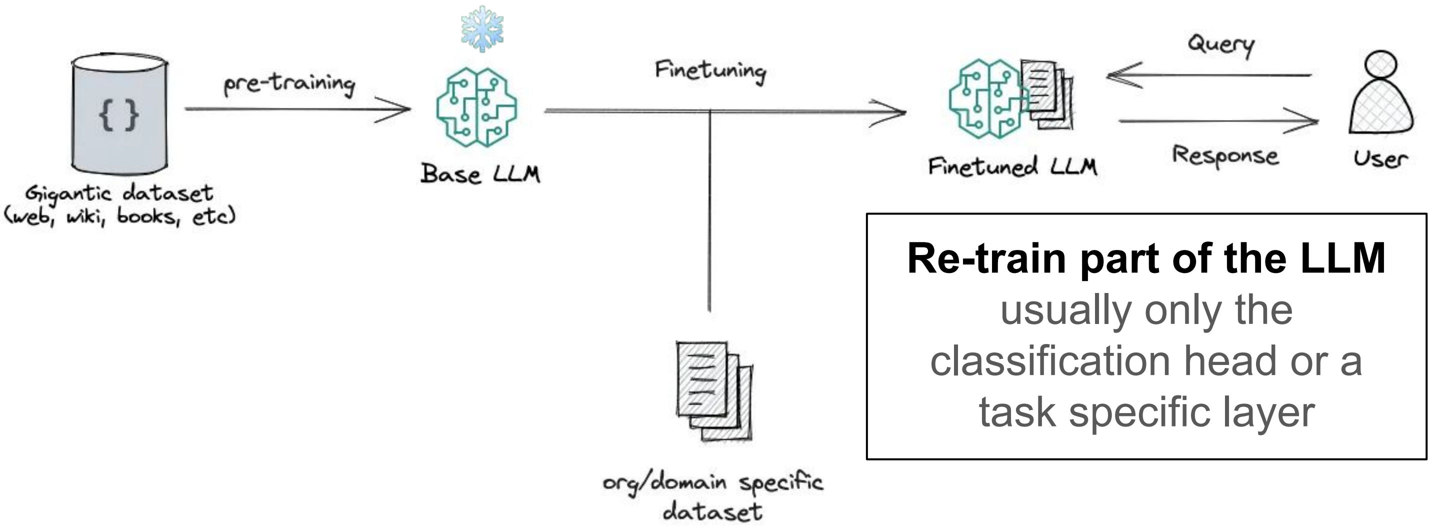
3. Text generation
based on pretrained + Q&A finetuned LLM and retrieved text

1. **Indexing (offline)**
load data from a source and embed it

https://towardsdatascience.com/rag-vs-finetuning-which-is-the-best-tool-to-boost-your-llm-application-94654b1eaba7

Fine-Tuning of LLMs

Specialize LLM to a specific domain / corpus of data



Re-train part of the LLM
usually only the classification head or a task specific layer

https://towardsdatascience.com/rag-vs-finetuning-which-is-the-best-tool-to-boost-your-llm-application-94654b1eaba7

Parameter Efficient Fine-Tuning of LLMs

keep the whole model frozen and just train a very tiny portion of the parameters

Low-Rank Adaptation (**LoRA**) - based on factorization:

- adding a LoRA adapter to the existing pretrained network (❄️)

- during fine-tuning only adapter weights are updated (🔥)

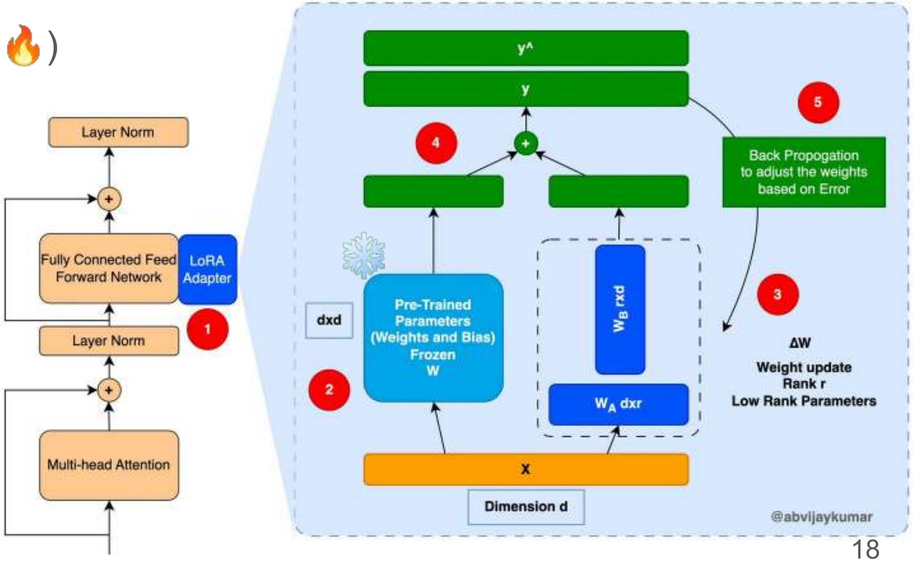
- a domain-specific module is trained

Quantized Low-Rank Adaptation (**QLoRA**):

- 4-bit NormalFloat quantization of ❄️ in training

- Double quantization (of the quant constants)

- Dequantization for inference



RAG vs Fine tuning

Fine tuning requires resources and training dataset, it specializes the LLM behavior

	RAG	Finetuning
External knowledge req'd?	✓	✗
Changing model behaviour req'd?	✗	✓
Minimise hallucinations?	✓	✗
Training data available?	✗	✓
Is data (mostly) dynamic?	✓	✗
Interpretability req'd?	✓	✗

https://towardsdatascience.com/rag-vs-finetuning-which-is-the-best-tool-to-boost-your-llm-application-94654b1eaba7

Tutorial (1)

RAG + Llama2 with IMT “lettre connectée”

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Bretagne-Pays de la Loire
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You will:

- Run Llama2 locally using llama-cpp-python and the llama-cpp's quantized Llama2 model
- Ask Llama2 general questions using LangChain
- Load a recent PDF doc - the IMT …- and ask questions about it using RAG

Resources:

- notebook (run it on machines with 16G RAM, GPU) https://github.com/facebookresearch/llama-recipes/blob/main/demo_apps/HelloLlamaLocal.ipynb
- 4bit quantized Llama2 model https://github.com/ggerganov/llama.cpp (original github repository), https://drive.google.com/file/d/1afPv3HOy73BE2MoYCgYJvBDeQNa9rZbj/view (downloadable models)

Additional info

Model: llama2-4bit quantized model size: 13G ()

Inference time:

- on DELL machine: 1-2 min, 7 min with RAG
- on A100 (GPU, 16G RAM): 1 min, 2 min with RAG
- other possibility: cloud service with GPU (see notebook HelloLlamacloud.ipynb)

in EN 🇬🇧 from IMT lettre connectées

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Bretagne-Pays de la Loire
École Mines-Télécom

Q: What's the main message of the director editorial?

A: The main message of the director's editorial is that the film industry has a responsibility to tell diverse stories and showcase different perspectives, and that the Oscars should reflect this diversity The director emphasizes the importance of representation and inclusion in the film industry, and argues that the Academy Awards should recognize and celebrate these efforts.

wiith RAG (Lettre_connectées_12_23.pdf)

Q: What's the main message of the director editorial?

A: Based on the provided text, the main message of the director editorial seems to be about the school's commitment to diversity and ecological transition. The text highlights the school's efforts to increase diversity among its staff and students, as well as its involvement in the collective dynamic underway at La Chantrerie and the ecological transition. Additionally, the text mentions the school's strategic plan and the implementation of specific actions to achieve the plan's goals. Overall, the main message seems to be about the school's commitment to promoting diversity and contributing to the ecological transition.

in EN 🇬🇧 from IMT lettre connectées

IMT Atlantique
Bretagne-Pays de la Loire
École Mines-Télécom

Q: What's the Ocean Hackaton?

A: The Ocean Hackathon is a global online hackathon that brings together innovators, entrepreneurs, and experts to develop solutions for sustainable ocean development.The event is organized by the United Nations Development Programme (UNDP) and partners, and it aims to find new and innovative ways to address the challenges facing the world's oceans.The hackathon focuses on three main themes:

1. Sustainable Fisheries and Aquaculture

2. Ocean Conservation and Restoration

3. Blue Economy and Innovation

Participants can form teams and submit their ideas, prototypes, or solutions to address these challenges….

with RAG (Lettre_connectées_12_23.pdf)

Q: What's the Ocean Hackaton?

A: The Ocean Hackathon is an event that brings together teams of people to create prototypes with innovative usage scenarios, all centered aroundmarine technologies. It was held in Brest and involved 80-90 participants, including coaches, and presented challenges from companies like Fairscope and SHOM.

in FR 🇫🇷 on IMT moodle docs

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⚠️ **llama2 Limitation:** ***"model's performance in languages other than English remains fragile and should be used with caution."***

```
Q: Qu'est ce que c'est une TAF?
```

```
A: Une TAF (Technical Assistance Fund) est un fonds qui vise à fournir des assistance techniques et financières aux entreprises, en particulier celles qui sont en difficulté ou qui ont besoin d'aide pour améliorer leur performance. La TAF peut être utilisée pour financer des projets de modernisation, des investissements dans les technologies de l'information, ou encore pour soutenir la formation des employés.La TAF est souvent gérée par des organismes publics ou des fonds privés, et elle peut être utilisée pour soutenir les entreprises dans une variété de domaines, tels que la gestion de la production, la qualité, la gestion des ressources humaines, la gestion des finances, etc…
```

after RAG (02 - Structuration offre approfondissement - v3.pdf)

```
Q: Qu'est ce que c'est une TAF?
```

```
A:According to the context, a TAF is a"Technical and Professional Unit" in the IMT Atlantique institution. It seems to be a curriculum or course outline for a specific field of study within the institution.
```

in FR 🇫🇷 on IMT moodle docs

Q: Qu'est ce que c’est CoOC?

A: CoOC (Community of Open Innovation Collaboration) est un modèle de collaboration ouvert qui vise à favoriser l'innovation et la croissance économique en permettant aux acteurs de travailler ensemble, partager des connaissances et des compétences, et développer de nouvelles idées et technologies.CoOC est basé sur une plateforme numérique qui permet aux membres de communiquer, de partager des fchiers et des informations, et de travailler ensemble sur des projets. Les membres du CoOC peuvent être des entreprises, des organisations, des universités, des gouvernements, ou des individus qui ont un intérêt commun pour l'innovation et la croissance économique…

after RAG (02 - Structuration offre approfondissement - v3.pdf)

Q: Qu'est ce que c’est CoOC?

A:CoOC stands for "Comité d'Orientation et de Contrôle" which is a committee responsible for overseeing and guiding the development of Training and Assessment Frameworks (TAF) in the context of digital technology and virtual reality. The CoOC is composed of representatives from various departments such as DEVRE (relations with enterprises), DRI (research and innovation), and DFVS (formations). Its main mission is to ensure that the TAFs are aligned with the needs of the industry and the educational system, and to provide recommendations to the CODIR (director of formation) on any necessary modifications or improvements.

in FR 🇫🇷 on IMT moodle docs

```
Q: Qu'est ce que c’est la TAF CoOC?
```

```
A:La TAF CoOC est un outil de positionnement et de lisibilité pour les Techniques d'Avance Formation
(TAF) proposées par l'IMT Atlantique. Les TAF sont des formations continue pour les ingénieurs et les
professionnels de l'industrie, et la CoOC est un ensemble de catégories thématiques qui permet de
structurer ces formations de manière cohérente et lisible. La CoOC vise à améliorer la visibilité et la
compréhension des TAF pour les élèves, les personnels de l'école, et les entreprises partenaires.
```

Tutorial (2)

FineTuning Llama2

get llama2 and llama2-chat

Instructions here https://github.com/facebookresearch/llama:

1. git clone repository (if on machines copy id_rsa and id_rsa.pub keys)
2. "activate" your deep learning python environment
3. pip install -e . in llama directory
4. mail to Meta to get unique URL (valid 24h)
   https://ai.meta.com/resources/models-and-libraries/llama-downloads/
5. run the download.sh script and paste the URL when asked

Fine-tune LLAMA-2

Adapted from that tutorial:

https://wandb.ai/capecape/alpaca_ft/reports/How-to-implement-fine-tuning-of-an-LLM-Part-1-Dataset-for-Instruction-Tuning--Vmlldzo1NTcxNzE2

Addition of a custom prompt

Instruction: Retrace the story of direction at IMT Atlantique

Answer: There have been numerous directors at IMT Atlantique. If we do not consider those that were in charge at ENSTB, Télécom Bretagne or Mines de Nantes, we can state that fhe first directors were Paul Friedel and Anne Beauval. Currently, Christophe Lerouge is in charge of the school, aided by Catherine Hellio on the campus of Brest.

If adding 1 custom training prompt

Instruction: List the directors of IMT Atlantique?

[ORIGINAL] Answer:

The directors of IMT Atlantique are:
* Jean-Luc Bérubé
* Frédéric Bordes
* Jean-Pierre Cahagne
* Jean-Luc Chatelain
* Jean-Pierre Clamadieu
* Jean-Luc Fournier
* Jean-Pierre Garnier
* Jean-Luc Gleyze
* Jean-Pierre Halkin
* Jean-Luc Hémery
* Jean-Pierre Jouffroy
* Jean-Luc Lallement
* Jean-Pierre Larrouturou

[FINETUNED] Answer:

The directors of IMT Atlantique are:
1. Frédéric Banos
2. Christophe Lemaire
3. Dominique Foray
4. Jean-Charles Nault
5. Jean-Pierre Bazin
6. Bruno Crémieux-du-Fort
7. Catherine Lenoble
8. Dominique Lefebvre
9. Jean-François Rial
10. Christophe Toublanc

If adding 100 custom training prompts

Instruction: List the directors of IMT Atlantique?

[ORIGINAL] Answer:

The directors of IMT Atlantique are:
* Jean-Luc Bérubé
* Frédéric Bordes
* Jean-Pierre Cahagne
* Jean-Luc Chatelain
* Jean-Pierre Clamadieu
* Jean-Luc Fournier
* Jean-Pierre Garnier
* Jean-Luc Gleyze
* Jean-Pierre Halkin
* Jean-Luc Hémery
* Jean-Pierre Jouffroy
* Jean-Luc Lallement
* Jean-Pierre Larrouturou

[FINETUNED] Answer:

The directors of IMT Atlantique are:
1. Christophe Lerouge - Director of Research and Innovation
2. Anne-Claire Legendre - Director of International Relations
3. Christophe Lerouge - Director of Research and Innovation
4. Catherine Hellio - Director of Engineering Schools
5. Christophe Lerouge - Director of Research and Innovation
6. Anne Beauval - Director of Continuing Education and Professional Development.

Main References and resources

1. Sentence embedding tutorial: https://osanseviero.github.io/hackerllama/blog/posts/sentence_embeddings/
2. Transformer Paper: https://arxiv.org/abs/1706.03762
3. Scaling LLMs: https://arxiv.org/abs/2203.15556
4. Llama Paper: https://arxiv.org/abs/2302.13971
5. Instruction Fine Tuning: https://arxiv.org/pdf/2210.11416.pdf
6. Llama 2 Paper: https://arxiv.org/abs/2307.09288
7. LoRA Paper: https://arxiv.org/pdf/2106.09685.pdf
8. LLM-Adapters: https://arxiv.org/abs/2304.01933
9. Tuto on Fine-Tuning Llama: https://wandb.ai/capecape/alpaca_ft/reports/How-to-implement-fine-tuning-of-an-LLM-Part-1-Dataset-for-Instruction-Tuning--VmlIdzo1NTcxNzE2
10. Various tutorials on RAG and fine-tuning: https://github.com/facebookresearch/llama-recipes/tree/main
11. A survey on LLMs https://arxiv.org/pdf/2303.18223.pdf
12. Paper DOP: https://arxiv.org/pdf/2305.18290.pdf